

Quality assessment of RNA sequencing data after KRAS suppression

WEEK 2 PROGRESS REPORT

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Background and knowledge gap

RNA sequencing can reveal responses to KRAS small interfering RNA (siRNA) in pancreatic ductal adenocarcinoma (PDAC). Reliable comparisons require intact inputs, suitable read quality and correct sample matching.

Objective

Assess read quality and prepare the paired files for reference-genome alignment.

Methods

We checked European Nucleotide Archive (ENA) sample records and 34 FastQC 0.12.1 reports: 32 from Galaxy and two from complete local recovery files. MultiQC 1.35 combined them. The 17 paired-end runs represent eight cell lines, with an extra MiaPaca-2 KRAS run of unresolved repeat type. HISAT2 2.2.3 aligned the first 1,000,000 pairs of Control M1 and KRAS P1 to human hg38 under unstranded, forward-stranded (FR) and reverse-stranded (RF) settings.

FastQC checks across 34 files

Check	Pass	Warn	Fail	Files
Per-base quality	34	0	0	34
Per-tile quality	34	0	0	34
Per-read quality	34	0	0	34
Base composition	0	0	34	34
GC distribution	32	2	0	34
N content	34	0	0	34
Read length	34	0	0	34
Duplication	0	0	34	34
Overrepresented sequences	34	0	0	34
Adapters	34	0	0	34

Results

All 34 files contain 150-base reads, with 37,886,040–47,840,900 reads per file and 50%–52% GC. Mate counts match across 17 runs. MultiQC confirms 34 entries and 57.0%–67.3% duplication. Both SRR24828471 mates have GC-distribution warnings; contamination is not established.

Alignment pilots and interpretation

All six pilots completed. Overall read alignment was 96.63% for Control M1 and 96.40% for KRAS P1 under every strand setting. Concordant unique-pair alignment was 86.40% (unstranded) or 86.44% (FR/RF) for Control M1 and 87.43% throughout for KRAS P1. These non-random subsets are not full-run results. The samples are different cell lines, so their difference cannot be attributed to treatment. Identical FR/RF rates do not establish library orientation.

Quality interpretation and repairs

RNA library preparation and unequal transcript abundance can cause composition and duplication failures; these labels alone do not justify trimming or duplicate removal. Truncated imports were excluded, not deleted. Complete local files restored QC coverage. SRA recovery also restored both Galaxy inputs; independent FastQC confirmed matching full-length counts and module flags. Next, establish strand specificity and complete full-run alignments. Compare treatments within cell lines. No gene-count or differential-expression results are claimed.

References

1. Galaxy FastQC, MultiQC 165, HISAT2 summaries 122–136 and local recovery QC; ENA PRJNA980201. Checked 22 September 2026. Records: github.com/yichuanlin07/pdac-kras-rnaseq.
2. Babraham Bioinformatics. FastQC documentation: Per Base Sequence Content, Per Sequence GC Content and Duplicate Sequences. bioinformatics.babraham.ac.uk/projects/fastqc/
3. Klomp JA et al. Defining the KRAS- and ERK-dependent transcriptome in KRAS-mutant cancers. Science. 2024. doi:10.1126/science.adk0775.

Author contributions

Yichuan Lin organized the data and coordinated the analysis and preparation of this report.

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